Ensuring LOWER THERMAL DISSIPATION For Higher Efficiency

Vertica Asthana is a technical journalist at EFY

A renowned manufacturing firm incurred heavy monetary losses when clients returned 5000 units of its LED lights after discovering that the supplied units failed within a few hours of usage.

On investigation, development engineers at the firm found heat dissipation issues with MOSFET and SMPS, which were not taken care of during the design stage. The excessive heat led to the LEDs' failure.

Heat dissipation causes

The design of a system, coupled with judicial use of components, is crucial as it determines the product's efficiency, and, in turn, thermal dissipation.

Pratik Mukherjee, vice-president (technology), SAR Group, recalls an incident where they had designed a sine-wave inverter with a totally different topology from other inverters. As the design was very sound, the company reaped its benefits and continues to do so even today. Although Mukherjee had to face a lot of criticism for the long time taken for its research and development, the product turned out to be a milestone for the company.

System over-usage and several applications running simultaneously also affect thermal dissipation. In a multi-functional design, several components of a system work simultaneously, which not only affects the other components but also increases heat dissipation. Components fail when these operate at temperatures exceeding the thermal level or junction temperature threshold.

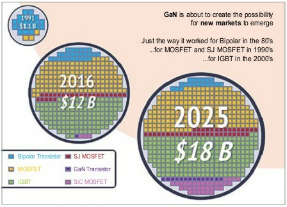

Fig. 1: GaN MOSFET (Image courtesy: www.pntpower.com)

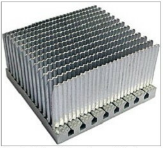

Fig. 2: Skive, Zipper heat-sink (Image courtesy: www.wakefield-vette.com)

Upcoming innovations

PCB manufacturers have been looking at various ways to reduce heat dissipation from their PCB boards, and make the end products efficient as well as durable.

To reduce thermal dissipation, a few upgrades have been introduced—from power supply and PCB routing to transformer and MOSFET manufacturing, and from solar cells in a panel to batteries in vehicles.

Power supply. Hrishikesh Kamat, CEO, Shalaka Technologies, says, "The design consideration, including the placement and calculative use of components, has a huge impact on thermal dissipation."

Smartphone makers such as Micromax are ditching the traditional way of reducing thermal dissipation. For instance, in a mobile charger, a small and compact device, the voltage step-down process is followed by rectification to substantially reduce heat dissipation. Traditionally, the rectification process preceded voltage step-down in capacitive circuits, thus generating a larger amount of heat.

Kamat informs, "Most smartphone makers are using this technique to bypass heat issues, while a few are resorting to the use of ICs."

Conor Quinn, technical marketing senior director, Artesyn Embedded Technologies, says, "Many of today's power supplies use switching devices like MOSFETs or gate drivers, which are supplied in IC-style packages. Most ICs have in-built thermal protection and cut-off at a certain temperature before it hits the junction point."